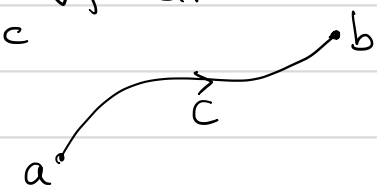
Applications to electromagnetism

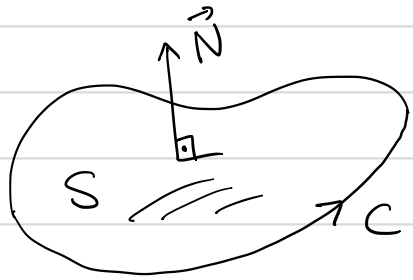
Fundamental theorem (FTC):

$$f(b) - f(a) = \int_C \nabla f \cdot d\vec{r}$$



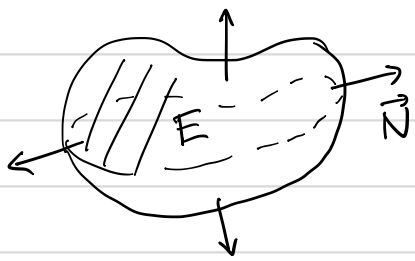
Stokes' theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$



Divergence theorem

$$\oiint_S \vec{F} \cdot d\vec{S} = \iiint_E \operatorname{div}(\vec{F})\, dV$$



Electromagnetism

Lorentz force: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

Maxwell's equations:

$$\nabla \cdot \vec{E} = \rho / \varepsilon_0$$
$$\nabla \times \vec{E} = -\partial \vec{B} / \partial t$$
$$\nabla \cdot \vec{B} = 0$$
$$c^2 \nabla \times \vec{B} = \frac{\partial \vec{E}}{\partial t} + \vec{j} / \varepsilon_0$$

Gauss' law:

$$\iint_S \vec{E} \cdot d\vec{S} = \iiint_E \overbrace{\nabla \cdot \vec{E}}^{\operatorname{div}(\vec{E})}\, dV$$

$$= \frac{1}{\varepsilon_0} \underbrace{\iiint_E \rho\, dV}_{\text{charge inside}} =: Q$$

$\rightsquigarrow$ Capacitors, ...

Inverse-square law:

For a point-charge at the origin: $\vec{E} = E(|\vec{r}|) \dfrac{\vec{r}}{|\vec{r}|}$

(by symmetry)

S = sphere of radius d about the origin

$$\frac{Q}{\varepsilon_0} = \iint_S \vec{E}\cdot\hat{N}\, dS$$

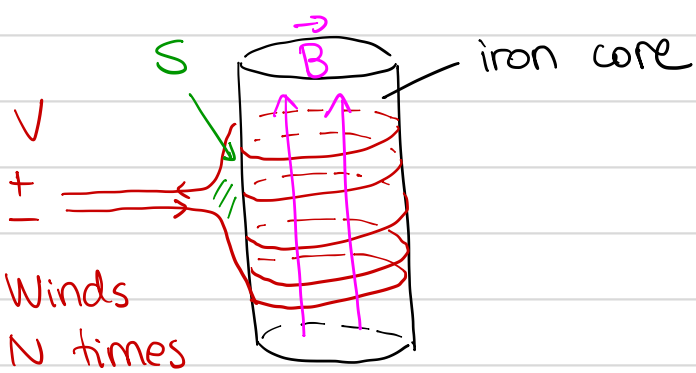
$$= \iint_S E\,\frac{\vec{r}}{|\vec{r}|}\cdot\frac{\vec{r}}{|\vec{r}|}\, dS$$

$$= \iint_S E\,\frac{\cancel{\vec{r}\cdot\vec{r}}}{\cancel{|\vec{r}|^2}}\, dS$$

$$= E\cdot\underbrace{\text{Area}(S)}_{4\pi d^2}$$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{d^2}$$

Faraday's law of induction



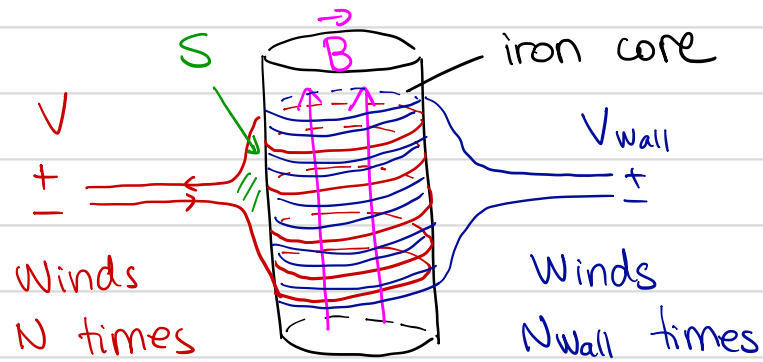
$$V = \int_C \vec{E}\cdot d\vec{r}$$

$$= \iint_S \nabla\times\vec{E}\cdot d\vec{S}$$

$$= \iint_S -\frac{\partial\vec{B}}{\partial t}\cdot dS$$

$$= -\frac{d}{dt}\iint_S \vec{B}\cdot d\vec{S} = -\frac{d}{dt}\, B\cdot N\underbrace{A}_{\text{area of cross-section}}$$

Transformers



$$\frac{V}{N} = -\frac{d}{dt}B\cdot A = \frac{V_{wall}}{N_{wall}}$$

Example: $\frac{110\text{ V}}{5\text{ turns}} = \frac{220\text{ V}}{10\text{ turns}}$.